

ActInf Livestream #049.1 ~ "A Worked Example of the Bayesian Mechanics"

Livestream | 1:58:52

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And we're really appreciative, Dalton, that you've joined today. Looking forward to how these discussions go. So we can begin just by saying hello and anything that we want to explore or discuss today.

And then we'll just jump right into it. So I'm Daniel. I'm a researcher in California. And I think some of the questions that are really motivating me today is,

what is mechanical about Bayesian mechanics? And how do we take concepts that have physical interpretations like center of gravity, or even some of these more atomic level terms that were brought into play?

How do we take concepts that have physical

and how do we take concepts that have physical learning? I think it is? And personally I had immersed myself for the past several months in Dalton's paper.

And for the past couple of months we had fascinating discussions around this particular paper. So I'm feeling kind of star struck right now.

this particular paper. So I'm feeling kind of starstruck right now. And I'm very much looking forward to our discussion. And I'll pass it to Jakob.

Hello, I'm Jakob. I'm a student in the UK. And I suppose, I'm also very, very excited to be here today and to discuss the different implications of Bayesian mechanics and how it's related to other areas in contemporary physics and how that can then be applied to cognitive sciences and neuroscience. Awesome. So welcome. Thanks again, Dalton. Feel free to just begin where you'd like and bring us to the paper.

Of course. Yeah. I guess I'll also introduce myself just in case there are people in the audience that don't recognize me or have a face to a name. I'm Dalton. I'm a researcher. I'm based at Stony Brook University in New York, where I spend most of my time in the Department of Mathematics and the Department of Physics and Astronomy. I'm also at the Versys Lab, where I'm on the Mathematics and the Physics of the Free Energy Principle. So it's a pleasure to meet all three of you. And thank you, of course, for the kind words. I think we can maybe begin with a very broad overview of the paper, I will say. So one of my one of the aims of my research program is to get a better understanding of the mathematics of random dynamical systems, especially coupled random dynamical systems or systems that model non-equilibrium phenomena. And one of the interesting things to me

is the idea that within cognitive science or neuroscience, there are some algorithms that have been developed over time that have been built specifically to cope with these kinds of problems, which are problems that we don't know really how to make sense of mathematically, even physically. So we're on the back foot. But in neuroscience, the primary object of study is a non-equilibrium system. It is non-equilibrium and understanding non-equilibrium phenomena in the brain. And so one supposes that the methods that have been developed over time in this area, just maybe by trial and error, or at least trial and error by the standards of perhaps a pure mathematician, may have some insights into maths and physics in the same way that there's a very rich history of physics, inspiring more advanced mathematics, and then mathematics circling back and making physics more rigorous. I think there's a great opportunity to do that in biophysical systems. And I think the free energy principle is a source of great inspiration along those lines. So with that meta commentary out of the way, one of the things that I wanted to do with this paper is to actually drill down on that point. Because I think it's something that maybe gets lost in translation. If you just read kind of traditional active inference papers that are all about modelling cognition, I think there is there's something on the other side of the aisle to be said about what do those models of cognition say about modelling systems that are cognitive? Or what does it say about complex systems that are cognitive-like? And how can we make sense of those things mathematically? So that's in some sense the motivation for Bayesian mechanics. And that goes back to, well it goes back a long time, it goes back to probably Karl himself. He mentions something like this at a couple of points in the 2019 monograph. But it was worth writing a paper specifically with the motivation that if we consider what's been written about the free energy principle so far, there's a very nice foundation for better understanding the mathematics and physics of coupled or non-equilibrium stochastic processes. And so that was the motivation for this particular paper. And in addition to building out the fundamental mathematics in the first few sections, it's then possible to contextualize all that work with a kind of nice worked example of what does this mean for some kind of simple systems? How can we actually take this foundational maths and make it algorithmic again? And then foreshadowing maybe even greater complexity in the very last section. So yes, a lot of this goes back to a paper that I co-authored with a number of other people earlier in the summer called on Bayesian mechanics, where it's proposed that, you know, complex systems have a kind of informational physics about them. And that maybe the material physics or the maths for these systems is quite challenging in general. But maybe if you map these systems into the informational world, you can utilize new techniques and make sense of these kinds of systems by

approaching it from a different kind of viewpoint. That is, I think, the strength of Bayesian mechanics, as again, Carl and others have construed it, is it is a way of talking about informational physics, or the physics of Bayesian beliefs, the laws of motion on a statistical manifold. This is various ways of kind of couching that in.

more formal language. But again, that was the motivation for the paper. And then, as that gets developed, eventually, there's a point where we really get into the trenches and actually try and deploy this in some useful way. I think that those are really the kind of two aims that come together in some sense.

Thank you for the summary. While there's many ways to begin and start, perhaps we could start with the context that we brought,

which was thinking about this, and you touched on some of these threads as well, such as the key literature and some of the history.

Let's continue on in this context setting before we get to the formalisms and the worked examples and talk about the three kinds of theories, including what you mean by a theory, and then the three non-Bayesian

mechanics. And that'll set us up really well to see where Bayesian mechanics comes into play.

So this was in that paper just mentioned, on Bayesian mechanics, a physics of and by beliefs.

So what are dynamics, mechanics, and principles? And how were you using theory? And why was it so important

to be clear about what a theory was and the distinctions amongst these three terms?

Yeah, of course. So in that paper, there is this kind of trichotomy that gets introduced, where we talk both about dynamics and mechanics and principles and disentangle them in a very particular way.

So we think about dynamics as a kind of observational thing. And it's important to separate that from mechanics,

because complex systems as a science is very interested in dynamics. Can I kind of infer the laws of these kinds of systems just by modeling them using some kind of curve fitting or using some kind of set of canonical models that I know sort of fits this situation? But at the end of the day, yes, dynamics are mostly about the particular trajectories that a system takes through state space or physical space.

And this is conflated sometimes with what I call mechanics in that paper, which is something a little bit different. It is the laws from which these trajectories are determined. So for instance, dynamical systems people, especially in pure maths, really like the example of billiards. So you send one billiard

ball crashing into another one, and you could talk about the trajectory that the ball takes. And you could write down the equations of motion for this system and just add up all the forces acting on it and figure out what direction is this ball that got collided with, what direction is that going to go into, with what momentum, and all of these things. But then figuring out that trajectory, once you plug in all of these kind of boundary conditions, the mass of the ball, the angle of the initial contact, and so forth, the way that you can kind of compose all that information together, that needs to come from

somewhere. And that's what we call, or what I call a mechanical principle. And this is not a distinction that is new to the literature. I'm not breaking new ground here, but I think it is one of those things that's not made sufficiently clear in the FEP literature, or at least in the physical, the places where it's adjacent to physics. So mechanics are about the laws that give you those trajectories. So we know that the force is acting on one or another billiard balls summed together because of Newton's law of motion. That's a mechanical theory. If I have F , then I get M times A , and I can plug in F and I can plug in A and I can get a trajectory. But if I abstract away from those particulars, I get some kind of law that tells me, okay, how a trajectory is generated in the physical world.

Abstracting one more step, we can then ask, okay, so where do mechanical theories come from? What tells me that I have the right mechanical theory generating the right dynamics? And this is about the kinds of very fundamental laws that we know hold everywhere in the universe at a given scale. So these are principles like conservation of energy, and this gives you the behavior of collisions or the principle of stationary action, which actually gives you back Newton's law. So you can now chain these things together and say, you know, one ball hits the other one, it goes in this direction, because if you sum the forces and you add up all the directions, you get a net movement. And you can do that because these billiard balls minimize or at least make stationary their action. So their motion must follow this particular mechanical law, and that's the law that generates these trajectories. So the reason why this is mentioned in the Bayesian mechanics paper is because these have, I think, been conflated in the literature previously, because you have the idea of the free energy principle, which is a principle, it's a thing that

gives you back some kind of generative rule for the dynamics of specific systems. And that is usually something to do with the motion of internal states on a statistical manifold, in the sense of, for instance, Thomas Parr, Lance de Carsten, and Carl Fristen's article, Stochastic Thermodynamics, Markov Blankets, and Information Geometry. I think that's the title anyway. So if you say, okay, the dynamics of a system whose parameter performs approximate Bayesian inference minimizes the surprisal, then you get approximate

Bayesian inference as a law of motion or the dynamics of some object. And this law is usually written as a particular gradient flow relation on the space of beliefs. So it tells you how do beliefs change as those parameters doing the inference change. And then if you add in progressively more details, okay, what do those internal states actually mean in the context of the system? What are those beliefs about? These are the kind of boundary conditions that give you the trajectories, the specific dynamics of the system. And there's an interesting sense in which these boundary conditions are literal boundary conditions because understanding what's internal and what's external and being inferred is precisely the specification of a Markov blanket. So as soon as you actually give the system a very particular partition, as soon as you carve it up in a specific way, you start to get a sense of the actual evolution of a system as one might see it in the real world. But this is, I think, separating things out in a formal sense like this is useful because it avoids skipping from step one, which is minimize variational free energy or surprisal. And then going straight away to this is how something like an active inference system does

prediction and action. You know, there's some steps in the middle that haven't really been filled out if you just go from A to B directly. And when you begin to talk about the maths and the physics, collapsing things together like that

starts to throw away some important details that actually do make a difference in the way that you tell the story.

Awesome. Ali or Jakob, anything you want to add in or ask there?

Yeah, Jakob.

Yeah, I would maybe ask regarding the kind of laws of motion of Bayesian mechanics. You mentioned approximate Bayesian inference.

Would that imply that the 3D faces of Bayesian mechanics, the mode matching, mode tracking and path tracking,

would you say those are also specific laws of motion within Bayesian mechanics?

You could say that, yeah. I think at this point the distinction gets very fine. And so there is a place to usefully truncate it. But yeah, these are three different kinds of approximate Bayesian inference.

And so they are three different kinds of motion under free energy principle minimization.

And so this is kind of, I guess, a middle area between dynamics and mechanics. You've started to specify some things like, am I doing approximate Bayesian inference over states or am I doing it over paths?

But you're still lacking actual dynamical details about what do those states and paths mean?

So this is maybe a middle area that blends the idea of approximate Bayesian inference as a law of motion under

surprise minimization and actually getting down to the very bottom of this pyramid. So if I were to put it somewhere into this figure, I'd maybe slot it in. Yeah, that's pretty much what I'm imagining is you have approximate Bayesian inference sort of roughly up here and then there are different expressions of that, just as there are different expressions of Newton's laws depending on if you're in fluids or rigid bodies or celestial mechanics. So you can introduce even finer partitions than these three.

There's many places to go. One question is what are these modes about?

Who or what is tracking what and when?

Yeah, so this is kind of about the idea being approximate Bayesian inference tells you that systems match their parameters in such a way that they perform inference, right? So when we think about Bayesian inference, we can talk about it as simply inferring the parameters of a probability density.

So in the free energy principle, when you get approximate Bayesian inference

saying, okay, systems that are coupled synchronize certain parameters, this is Bayesian inference.

And conversely, if you have that synchronization, you have Bayesian inference or you have approximate Bayesian

inference if you're inferring the parameters of an approximate density, which is where you get into

the factorization of the free energy functional that we are familiar with, where you have

a variational density matched to a joint density, but you can split it into the variational density

matched to a conditional density, plus an extra surprisal term that kind of measures how good that

approximation is. So this descends from that idea that you can deduce what a system is doing

just by saying, okay, it's performing approximate Bayesian inference, because that allows you to say its parameters are synchronizing to the parameters of the environment or of another system.

So that's what we mean when we say modes. In this case, a mode is just a parameter of the distribution, and it comes from some of Carl's earlier literature where he uses the Laplace approximation. So specifically, we're matching the, we're kind of fitting a Gaussian density to some arbitrary probability, and then the mode of that Gaussian is one of the parameters, because it tells you, okay, where is the center of that density. And if you also get a sense of where the variance of that density is, now suddenly, by fitting those two parameters, you can talk about the system, because now you can say, okay, the physical parameters of the system are encoding these parameters of some density. And in a very natural way, this gets you back much of statistical physics, because it's all about how does a system reflect these statistics of another system, probabilistically, how does this look. But this is kind of upgraded in Bayesian mechanics to noting that, well, that inference happens because the physical parameters of the system are matching the physical parameters of the environment. And that's where the probabilities come from. That's why we talk about beliefs. So when we say modes, we're referring to parameters that are fit for synchronization.

Excellent. Thank you. Jakob?

Since we have the schema on the slide there, I just wanted to ask, where do you think that the G theory formulation fits into this diagram? Because with G theory, there is approximate Bayesian inference, but it's not necessarily density over states. So is it another node directly from the Bayesian mechanics?

Yeah. So in this figure, the approximate Bayesian inference lemma is a very specific thing that originally goes back to, I believe, originally it's in a 2012 paper by Carl in Entropy called the Free Energy Principle for Biological Systems. But certainly it's a much bigger deal is made in the 2019 monograph. And it's the approximate Bayesian inference lemma, as he calls it, is specifically about mode matching. Approximate Bayesian inference is a bit more general. So if you were to kind of slot that in somewhere in here, it would maybe be parallel to Bayesian mechanics, because just like classical mechanics, the law of motion is Newton's laws. For Bayesian mechanics, the law of motion, the thing that generates specific physical dynamics, is approximate Bayesian inference. Or the law that when you do have synchronized systems, you can write it as variational free energy minimization in virtue of the parameters of the two systems matching up. And that's the parameter that minimizes the distance between a held density and an observed density. So approximate Bayesian inference lemma is that result specifically. But that's more general. The idea of approximate Bayesian inference is more general. And probably someone needs to write down a generalization of the approximate Bayesian inference lemma to paths specifically to talk about path tracking. But once you have a path that fits the bill as a mode, and once you have two systems evolving together tracking each other's modes or each other's paths, and if you have a probability density over those paths, you can now talk about approximate Bayesian inference

in the path setting. Now where that has to do with G-theory, so G-theory, the name and kind of the vision is actually due to Maxwell Ramstad, a colleague of mine, also here in the Versys lab. And he has suggested

that

G-theory kind of lies in the intersection between a few different things that fall out of the path-based formalism. So if you have approximate Bayesian inference over paths, if you're doing path tracking, then there are lots of things that do come out of this, almost canonically. One of them is everything that comes along with path integral formulations in quantum mechanics. And this is, I take advantage of this at various points in the paper. At one point I use it to explain, okay, why does Bayesian mechanics actually

make sense in the classical world? And then later on I use it to relate it to chaos. That's a specific example.

I think more generally path tracking deals with non-stationary systems very well, because you're allowed to talk about now,

how does the density, how does the belief change in time? How does the mode of a single belief change over time to make a kind of modal path? And then what are the marginals? If you slice up the density along that path, what do you get? So you can talk about non-stationary systems. You can talk about systems out of equilibrium

this way because you can relate the free energy principle over paths to what's called the principle of maximum caliber in statistical physics, which is all about non-equilibria and entry production and non-equilibrium steady states. So I think as we have written about it, G-theory is the

duality between path integrals and the FEP over paths, or the duality between max cal and maximum caliber and the FEP over paths. But I think that's selling it short a little bit because it's also

the thing that brings in all these other ideas. And stochastic thermodynamics is a really interesting one as well, right? Also falls out of the path-based formulas very well. Also does very well with

non-equilibria. So G-theory is kind of this larger object that makes sense of why that column of Bayesian mechanics actually works and explains a little of the true power of the path tracking as opposed to

mode tracking. Why are we interested in forgetting about states? And that's the explanation for that,

or the justification for that, resides in G-theory, which is supposed to explain why do all of these

things just come out of the path-based formalism? Why does it make sense? And why does the FEP over paths

lie at the center of a lot of disparate fields? Which I think is one of the real, very powerful statements

that may be possible in the near future. Is the FEP actually canonically the theory of everything in the sense

that there are lots of approaches in statistical physics and stochastic process theory that trace back to the FEP

or the FEP subsumes or implies or something like this. G-theory is supposed to kind of plug into that last column,

but that is somewhat outside of the hierarchy in that sense.

Excellent. Thank you. Ali?

Ali? Well, yeah, I was, when I was reading about G-theory, I thought it somehow relates to Rolf Landor's principle, which basically says that if an observer loses information about a system, the corresponding entropy in non-information bearing degrees of freedom of the information processing system or its environment is increased. Because, you know, Landor believed that information is physical. So he somehow

put it in terms of a kind of thermodynamical point of view. But how does it fit? Or can G-theory theory be viewed through the lens of Landor's principle, which I think is not a completely non-controversial principle as well?

Yeah, that's an interesting question. I think insofar as it does give you some technology to talk about entry production at non-equilibrium steady state. So it does give you a way of saying, okay, if I'm exchanging with my environment, if information is flowing out and flowing back in, then there is some increase in the entropy. And with respect to things like control theory, this is pretty interesting, because it allows you to talk about, okay, if the system is observing something about its environment, if it's constantly maintaining a steady state, so engaging in control, then you can talk about that entropically, and you can talk about the irreversibility of processes, even at steady state. And there are, you know, there are things that connect to that, like the Crookes fluctuation theorem, or Yarsinski equality, and things of that nature. So my guess would be, it probably covers something like Landau's principle in that context, talking about the energy required to erase a bit of information, say, is ultimately the energy required to insulate yourself from the environment. That minimizes your surprise. It means the environment does not offer information to you, but that does take energy, and sometimes quite a lot. So that's probably where that connects to Landau's principle, is talking about and quantifying the kind of cost of maintaining a Markov blanket, in probably entropic terms.

Landau's principle, is talking about the energy expenditure of the brain, compared to, say, a neural network operating on traditional computational architectures, using many, many light bulbs, many, many calories of energy, versus a brain, which is able to carry out some tasks with seemingly low energy expenditure, while other tasks it does not seem to be able to perform at all.

I want to pull back, before we go into the three non-Bayesian mechanics, and ask, what is Bayesian? What does it mean for something to be Bayesian?

Yeah, in my mind, Bayesian is about inference. It's about a particular method of doing inference that gives you consistent, unbiased results, and fits in very nicely with the rest of probability theory. In the context of Bayesian mechanics, it is the idea that the physics of material systems, or the material physics of complex systems, maps into a physics of beliefs. So you can talk about the parameters that are doing approximate Bayesian inference, or you can talk about approximate Bayesian inference. You can talk about, okay, what are the beliefs parameterized by that mode doing, and how does that reflect something about the dynamics of the system?

The reason why this is interesting is not just because it's a different approach to the problem, and dealing with interacting systems and systems out of equilibrium is an area that does need new approaches because it is a very poorly understood area within mathematics and physics. It still is, it has been for some time, I think.

But it's also interesting because complex systems are informational, so there's some sense in which this brings us closer to the metal. So it's not just a new view, it is the right new view, because complex systems are systems that compute and capture data.

They are systems that have memory. So it's not as simple as even, here's a new way of writing down the dynamics of random or stochastic processes based on the parameterization of some action function. That's valuable, because that's a new thing, but it's also, it fits very nicely into our understanding of complex systems as things that have an informational physics.

So, understanding the dynamics of things like the brain is contingent upon understanding what the brain is doing, and the brain is capturing and computing over data. It is storing memories, it is learning and sensing and inferring.

So, it's not just that it's a new view, it's really the right view, because it gets to the idea that complex systems do things physically,

because they are doing something in informational space. And the brain is a wonderful example of that. The brain is a coupled random dynamical system, it's coupled to its environment.

So you can talk about this parameterization thing, this synchronization thing, but it's even more insightful than just that, because the brain is doing approximate phase and inference.

So, I suppose it's no coincidence that the synchronization of coupled random dynamical systems gets you back this idea.

Because I think that is empirically just based on observations, I think that is as close to a physical law as one might be able to get at this stage in complex systems theory, is that things that synchronize do perform approximate phase and inference.

Because things that synchronize are things that are controlled systems or computational systems.

So, in that sense, I think it's right on the mark.

And as I said, you know, it's interesting that this goes way back to trying to figure out things about the brain. It goes way back to Carl Fristen's original work on generalized filtering and SPM and these kinds of algorithms that were designed to make sense of what is the brain doing physically based on what we think it's doing inferentially.

Right, the relationship between structural and functional connectivity and the relationship between neural representations and neural firing patterns.

But it has become something massively more general because it is the right kind of maths to talk about these sorts of systems.

And that's in turn because these sorts of systems do this kind of approximate Bayesian inference.

So, when you ask, yeah, what is Bayesian about Bayesian mechanics?

It is about the relationship between Bayesian beliefs and Bayesian inference and physical parameters representing something about the information available to that system or the statistics of a system coupled to that system.

Awesome.

Just a few notes before we then go into the non-Bayesian mechanics.

One is from ecological psychology, where many of our terms are drawn from, such as affordances. People have long pointed to the way that we use language like, can you grasp it, optimal grasp of ideas, and so tangible and peripersonal spatial ways of talking about ideas and cognitive processes.

And then, well, what is the first thing you want to know after you get optimal grasp?

How much does it weigh?

And you said, it has become massively more general.

Well, is that in grams or in what?

So, I see this as part of the development of the physicality of cognition in that the ecological psychology brought in relationality and action orientation, the pragmatic turn.

And then, now what about those objects?

If we drop the idea, what happens?

If we drop it from the discussion, what happens?

And then the second piece is the very interesting historical note about the development of complex systems and about the appropriateness of this kind of a formalism.

Because at the very least, standing as extremely on the instrumental spectrum as we can be, we're going to need to bake relationality into our analysis of any system, whatever it is that it is doing.

Absolutely.

So, great points.

Let us go to the non-Bayesian mechanics.

So, the non-Bayesian mechanics are classical, statistical, and quantum.

So, however you'd like, maybe just on each of the following three slides, just what are classical mechanics, statistical mechanics, and quantum mechanics kind of steel-personing the argument?

And just what do they explain?

What will they continue to be useful for?

And how can we see them as special cases or adjacencies or however you'd like to frame it with respect to Bayesian mechanics?

Sure.

Sure.

Yeah.

So, beginning with classical mechanics, I think is probably the simplest example.

You have kind of the physics of very large, very slow things.

Large is kind of a relative term, large compared to atoms maybe, and slow compared to the speed of light.

But nonetheless, this is a broad region of physics that describes things that don't have relativistic effects, things where observations commute.

So, it's non-quantum, things that are not noisy, so there's no probabilities, it's not statistical, and things that do not require relativistic corrections, so things that are below the speed of light.

When you have all four of those ingredients, I think I listed four anyway, you have classical physics, or at least you have a classical object, and the physics of that thing is classical mechanics.

So, this arises from, you may have inferred, four limits of other kinds of physics.

So, you want to be now infinitely precise, so this is the kind of no-noise limit of statistical mechanics, it's the slow speeds limit of special relativity, and it's a very interesting limit of quantum mechanics.

So, yes, but until then, I'll say, there is a classical limit of quantum mechanics, that you take a quantum field theory, and you can produce a classical theory out of that in a relatively well-defined way.

And again, I can talk a little bit of quantum mechanics, and again, I can talk a little about the nuances to that

statement.

There are some, but regardless, classical physics is about large things, really.

The interesting thing is that all four of these limits kind of coincide to some extent when you have a very large system.

So, very large systems allow you to use the law of large numbers to kind of sharpen your beliefs about a system.

So, you can imagine this as placing greater precision on the measurement of your system, and that recreates sort of the no-noise limit, because you get more and more precise in your estimate of the dynamics of the system.

Many body systems with very small objects are systems where you start to see quantum corrections.

So, particles, like atomic particles, for instance, you begin to run into a regime where things are small enough that quantum physics becomes relevant.

And also, large systems tend to move pretty slowly, as just a general rule, in what's called effective field theory, which talks about how one theory is a limit of another theory.

And in particular, it talks about as you increase in scale, you get simpler theories.

So, I think we would probably all agree classical physics is a lot simpler than quantum physics, and that's because effective field theory says that large things have simpler physics in very rigorous senses.

One expression of that is kind of the momentum scale is inverse to the mass scale.

So, larger things generally have lower momentum, lower energy.

Very small things whiz around at very fast speeds.

They have high momentum.

So, once you start to talk about large systems, things begin simplifying in such a way that more often than not, you run into classical physics as a sufficient description of the scale of observation that you're at.

So, classical physics, that's where it comes from.

It is about Newton's law.

Classical mechanics, Newtonian mechanics, as we might call it, is about Newton's law.

It says that, well, Newton's three laws, I suppose, I should say.

They say that every action has an equal and opposite reaction.

So, this is the idea that things that make, so contact forces have, at the classical level, an equivalent force in the opposite direction.

You can think of this as a kind of, if you're familiar with sort of exercises in school in this area, you're often asked to calculate the normal force of something.

The normal force is just what pushes up on me when I push down on something else.

So, this is why I don't phase through the ground when I stand up, is because there is an equal and opposite reaction.

I'm not only putting a force on the ground, but the ground is acting on me and keeping me up.

So, that's one of them.

The other one is the very famous $F = MA$.

It says that the force acting on a system is the mass times the acceleration of the system.

And likewise, you can deduce the acceleration of a system by taking the force acting on it and dividing out the mass.

So, this tells you, okay, how do we generate dynamics out of some boundary conditions like what is the mass, what direction is the force acting in.

And classical mechanics comes from a, what we call, least action principle.

And there are a number of equivalent ways of writing this.

But this generalizes to what some people call Lagrangian mechanics.

And that tells you, imagining what are the energies that those forces came from, how does the system behave with respect to its energies?

And if you talk about stationary action, you talk about minimizing energy, then you can recover Newton's laws as a system accelerates along force gradients.

And it doesn't waste energy by going any faster or going any slower.

It does exactly what it's told to do.

So, it uses precisely the energy budget available to it.

There's a very nice, well, maybe I shouldn't say it about my own paper, but there's a very lengthy exposition of this in the Physics of and by Beliefs paper that we've talked about once already.

I think it's section two, where I run through why is the analogy to classical mechanics even useful here?

And it's because of the stuff that we talked about previously with the very nice dichotomy or trichotomy between principles, mechanics and dynamics.

But anyway, more broadly than that, classical mechanics is a very good example of a least action principle.

Because it's the idea that the dynamics of classical things comes from a particular scale-appropriate limiting least action principle.

Excellent.

Ali or Jakob, anything to add or we'll move to statistical mechanics?

Okay.

Classical descriptions.

On to statistical.

Statistical.

Yeah.

So, statistical mechanics is a bit of an interesting one.

Statistical mechanics is a lot broader than I think one might initially think.

Because at the end of the day, it is to some extent just about systems with, as you've written here, probabilistic degrees of freedom.

And that covers a lot of things.

You know, you can, probability theory is just maths.

So, anywhere you can use probability theory to describe some system, you can use statistical mechanics.

And we know that this is, this generality does exist because you can talk about quantum statistical mechanics.

So, you can take the physics of very small, very fast things and use statistical mechanical methods.

You can use probability theory and entropy in these things.

And you can also go all the way up to black hole thermodynamics and you can do the same.

You can talk about, so black holes are very large things.

But you can still talk about the thermodynamics or statistical mechanics of black holes.

So, it's worth saying that we can talk about the no noise limit of a particular kind of statistical mechanics.

But then we need to be a little bit more precise about what that generates.

In this case, I'm thinking of classical statistical mechanics.

So, it's the statistical mechanics of interacting particles or systems with microstates that relate to macrostates via some kind of Gibbs relation.

And things that have free energy and, in particular, Shannon entropy.

So, this is a little bit more of a narrower idea of what statistical mechanics is.

Just because we need to constrain ourselves a little bit, even to get off the ground.

Otherwise, we'd be covering way too much.

But it is the case that if you take statistical descriptions of ensembles of objects,

and you become more and more precise and you get rid of the noise,

then you can reproduce kind of classical equations of motion.

And there are ways of doing that.

One such thing is called mean field theory.

So, in mean field theory, you get rid of the noise in a system.

And you just say, okay, here is the trajectory of the system in the same way as we might do with classical mechanics.

I'm skipping some details there.

And the analogy is a little bit more nuanced than that.

But it's kind of a useful mental picture, I think.

And one of the interesting things to note, I think, is what you've written here,

is that there is a principle that's relevant for statistical mechanics.

Because StatMech is very general, it's very mathematical.

It requires an equally mathematical principle.

And so, in probability theory, we have what's called the maximum entropy principle,

which is just a way of designing probability densities around inferences that satisfy some mathematical desiderata.

So, they should be consistent.

They should be unbiased.

They should lead back to Bayesian inferences under suitable conditions.

And Maxent is that thing.

And we know, due to, well, work by James, but more recently, work by people like Ken Dill, also here at Stony Brook,

that, and before him, mathematicians like Markowitz and Vilani,

all of this is a large body of work that points to the idea that if you maximize entropy, you get the mechanics of diffusion.

So, you get the mechanical rules that generate mass action, like diffusing particles.

Excellent.

I just want to ask about one word, which was unbiased.

So, what does unbiased mean here?

Unbiased, I think you can think of it as avoiding overfitting.

So, one of the things that you can consider Maxent to be is a recipe for doing inference that maximizes ignorance subject to some constraints.

So, if you have some known unknowns, then you want to incorporate those.

But if you have unknown unknowns, you don't want to risk overfitting for those things that you're not aware of.

So, if you're not aware of something, you don't want to falsely incorporate it into your inferences.

You could consider this as avoiding overfitting, or you could consider it as kind of optimizing for subclasses.

If you're thinking of probabilistic stuff as a problem of classification, then you can think of it as optimizing for subclasses that you don't have information on.

But either way, it is the idea that you don't want to introduce extraneous information into your inference, because that extraneous information biases you in a direction that the data doesn't support.

So, that's really the idea behind unbiased.

And one of the things that Janes did, and not only Janes, but also Shannon before him, who worked on the probabilistic aspects of this.

And then after Janes and Shannon, Shore and Johnson, who also were probability theorists and worked on Maxent axiomatically.

Not like Janes, who took it more as a physical rule for a long time, and then got into the probability theory of it.

Either way, all of these people are able to justify with rigorous proofs that maximum entropy is the best way of making unbiased inferences.

And there's lots of literature on this.

Those are three useful names, but there's a lot more due to Ariel Katicha.

I already mentioned Ken Dill, Adam Giffen.

Lots of people have taken an interest in this area.

Rightly so, I think.

Awesome.

We're going to keep moving to now quantum.

And then already many of the seeds that bring us to Bayesian are planted.

So, quantum.

What is it?

How did it come into play in your work?

Yeah.

So, quantum mechanics is, I think, qualitatively different from classical and statistical physics.

It's about very small things or very fast things.

And as I was saying, effective field theory tells us these things usually go together.

And it talks about kind of when you get things that are not classical.

Well, at least it started as when you get things that are not classical, what do you need to add to your theory to make the right predictions?

So, quantum mechanics began as a kind of modeling effort.

It grew from what we might call semi-classical physics, which is just about quantum corrections to classical stuff.

It grew from semi-classical physics to its own genuine branch of physics in and around the 20s, I suppose.

Certainly the early 1900s, anyway.

So, one of the things that becomes relevant is probability theory.

Because quantum mechanics, you may have heard about Heisenberg uncertainty.

It's about the fact that measurements do not commute.

So, the order that you do a measurement in matters.

A consequence of this is that there is always some inherent uncertainty when you measure so-called conjugate variables,

which are just things that fit into the theory as non-commuting pairs.

So, you measure two non-commuting pairs, there is some uncertainty.

You can measure one precisely, you can't measure both precisely.

And moreover, there's a trade-off.

If you measure one with infinite precision, you know absolutely nothing about the other one.

If you try and play half and half, then that's what you get.

You get kind of blanket uncertainty about both measurements.

So, because of the non-commuting of variables, and this is kind of an empirical fact, this doesn't come out of nowhere,

but it is one of those things where it just happens to be right because it's so elegant that it wouldn't make a lot of sense if it weren't.

Much like, I guess, the coincidence of synchronization and Bayesian inference in complex systems theory.

Because non-commuting variables gives us all of the rest of quantum mechanics, the stuff that we really know and love,

and also is a relatively well-defined way of getting back classical physics.

Because all you have to do is take the theory and make things commute, and then you get classical physics back.

I'm brushing some difficulties under the rug.

As you might imagine, it's not that simple.

And technically, it doesn't work, but it sort of does.

And if you're interested, I guess, you can look into what's called deformation quantization,

which is exactly the idea that if you take a classical system and you deform it in such a way that things no longer commute,

then you get back quantum physics.

But you can't really do it the other way around.

There's a bit of subtlety to the other direction.

Nonetheless, that is one route to making quantum stuff, classical stuff.

Another route is just focusing on the probabilities.

So because of this inherent uncertainty, quantum particles, their equations of motion, have what we call transition amplitudes, which gives the probability of moving from one state to another.

But again, nothing is certain in quantum mechanics.

If you, however, make it certain, mathematically, you can go in and just take the no-noise limit.

But more properly, you should do something called wick rotation first, which just allows that limit to make sense.

Although, well, yes, yes, you should do that first.

If you do that, you take the kind of no-noise limit, then you get certain state transitions.

And this just gives you back a classical dynamical system that you know where it's going once you have the equations of motion.

And this is the route that I take in the paper because it fits very nicely into the path integral formulation in a way that kind of just makes things make sense.

And one of the nice things that I then go on to say is if you have a parameter sitting inside that path integral, then you can make sense of why classical physics comes from the no-noise limit by writing it as a Bayesian mechanical problem.

You can write about synchronization and get back the Bayesian mechanics of classical stuff just by doing this very well-defined procedure.

Like, this is something that already exists in physics and is fairly uncontroversial.

So that's what you get in quantum mechanics.

It's about small stuff.

It's fundamentally probabilistic because there is fundamental uncertainty.

And again, this traces back to empirical facts.

I think that about covers it.

I don't know if I should have said more or if you have other questions.

I think that's sufficient at this benchmark for that slide.

Oh, it's great.

Jakob?

Jakob?

Yeah.

I realize the paper dealt mainly with Bayesian mechanics for classical objects, but do you see Bayesian mechanics being applied to non-classical phenomena like quantum tunneling or entanglement, which cannot really be explained in the limit of large numbers?

And then also a second question,

which is kind of on the intersection of quantum mechanics and statistical physics.

Is there any change in the formalism when you introduce the notion of indistinguishability between particles?

Sure.

Well, those are really good questions.

I think, yes, there is probably an idea of quantum tunneling that could be achieved, maybe, because quantum tunneling, if you're a physicist, this is something that you're familiar with on the nose.

If you're a mathematician, this is what you would think of as an instanton.

So instanton solutions to the equations of motion of a quantum field theory are tunneling dynamics.

So the idea becomes, okay, can we describe instanton solutions using Bayesian mechanics?

And the answer is I don't know, but it seems like it should maybe be possible, because there is at least an understanding of where points A and B are, if not how to get from A to B.

But that is something that would be interesting to do more work with.

And more generally, it would be interesting to think about quantum physics through the Bayesian mechanical lens, just because you do get this, you do recover this classical limit of the path integral.

And morally, path integral quantization should be equivalent to things like deformation quantization and geometric quantization and other methods even.

And actually, one other one is worth saying.

There's a thing called stochastic quantization, which begins from real valued probabilities, and then makes quantum stuff out of that.

And the Bayesian mechanics as it is construed today is very close to stochastic quantization in a few very key senses.

So one supposes it should be possible.

And that would be nice.

So that's the answer to one question, is should be, but I don't know.

It would be cool.

As far as entanglement and other kinds of information theory,

I actually don't know very much about this.

Quantum information theory is something that is relatively new, and a lot of people kind of in and adjacent to things like condensed matter theory

are very interested in this.

But I am behind on the literature, I think.

One of the things that does kind of lend itself to information theory in general is the fact that Bayesian mechanics has these connections to maxent and Bayesian inference. And these are, of course, ways of talking about information and information theory.

So, yes, my thought is something like entanglement, which has information theoretic undertones, should also be possible.

But again, I don't know initially what precisely that would look like.

There is some literature out there that has to do with the free energy principle in the context of information theory.

I am thinking of work due to Carl James Glazebrook and Chris Fields.

They've published a lot of work in this direction, which is also something I'm not totally familiar with.

But again, it seems like a very rich area to eventually pursue connections in.

Great. And then the second question was about indistinguishability of particles.

And then Ali.

Ah, yes. Okay.

Yeah. So I suppose you're thinking of passing to these kinds of bosonic field theories where you just pack a whole bunch of equivalent field states into your fox face and see if you can make that make sense.

I think you may notice this when we get into the supersymmetry results, is there is a conceptualization that, yes, things do change when you introduce bosonic looking things as opposed to fermionic looking things.

So it does become important to keep track of the statistics of these particles because that recovers things like BRST charges relating the two genres of particle.

And that is what gets the classical chaos in the last section.

So there is, yes, there is a reason to distinguish between the behavior of the particles in each theory.

I think the precise nature of that will become clearer once someone actually knuckles down and does out Bayesian quantum mechanics where we'll be able to couple directly to that kind of idea.

Awesome. A lot of fun topics.

The tunneling.

Do we have to visit every belief on the way from A to B?

Or, as it sometimes can seem, are we in one position and then we are in a different cognitive position?

All different kinds of experimenter, experimented questions in cognitive science, even potentially collective thought and ways of communication in terms of these formalisms.

Yeah.

And the instanton question is pretty interesting as well, right?

Because it talks about if we have a system with multiple classical minima.

So if we can say, okay, you know, there are multiple optimal Bayesian beliefs and instantons are the things that zip between those minima.

So it's a qualitatively different way of thinking about dynamical systems.

It's not a single minima and it's not a single classical trajectory.

It's multiple minima and it's almost like phase transitions between the two.

So certainly that's interesting from the point of view of physics, excuse me, from physics.

But tracing it back to something like cognitive science, thinking about, okay, what are the sort of phase transitions between our beliefs?

And can we actually access other phases, right?

Do those instanton solutions actually exist?

And do they describe some kind of zipping between maybe generative models or perceptory regimes or something of that nature?

It's very kind of nebulous, but it's a provocative question.

Excellent.

Oli?

Well, I was on the understanding that as we go from classical to statistical and ultimately to quantum mechanics, the role of information and uncertainty in describing and formulating the dynamics and mechanics of the physical systems becomes more explicit.

So is it a correct assumption to say that Bayesian mechanics somehow aims to put information back in the game and make it equally explicit through all of those mechanics?

Yeah, absolutely.

I think that's a good way of thinking about it, right?

The reason why Bayesian mechanics is so general, much like statistical mechanics is very general, is because it's just about information.

And lots of things have information, even at the classical level.

We just usually neglect it because we don't need to think about it.

But it becomes very apparent, and this is part of the explanatory power of doing something like that, it becomes very apparent that classical stuff comes about when that information goes away, or rather when those probabilities, that uncertainty goes away, that information becomes sort of limitingly large or infinite.

So what Bayesian mechanics allows you to do is actually keep track of how does information change between these different scales, which is one particular payoff that I think makes it very valuable.

But not just bridging scales, also applying to many different scales,
because lots of things do have information.

Information is just a mathematical apparatus that we can use to describe lots of things.

So if that's all that your physics is built on,
then it will be, by consequence, something very general.

Awesome.

The way that it's coming to me today is in classical mechanics,
we get, especially once we understand these limits,
which classical mechanics is like a point within a broader space of physics or mechanics,
we get principles of least action and stationarity and laws of large numbers,
not unlike the central limit theorem in statistics and a lot of the esoteric Gaussian statistics,
in statistical mechanics, the probabilistic nature becomes formal.

For example, in the billiards case in classical mechanics,
you might want to do statistics across many slightly similar billiards games.

And so at that point, you could either do the plug and chug frequentist statistics,
or at some point, you're going to want to think a little bit more broadly about the distribution of billiards.

And then quantum especially makes the relationality and the modeler modeled dialectic formal.

Because just by doing statistics, you're already in an area of explicit relationality,
map territory distinguishing.

And then quantum takes that to another level.

And could it be said that with Bayesian,
we're even going a layer deeper into the modeler's approach to the system?

Where do you see this cognitive turn?

I think so.

I think so.

The idea would be eventually we introduce an actual idea of information,
which naturally kind of depends on the observer or the modeler,
because information is kind of modeler dependent, right?

Information has a context.

It has a context.

It has a, or is contingent on observation.

So when we start talking about surprisal and the dynamics of beliefs and inference,
all of these are very explicitly putting the modeler back in the situation.

And that may be in the very metaphorical sense of a system synchronizing with its environment.

It doesn't necessarily have to be a cognitive observer or a panpsychist kind of modeler.

I don't really think there's remit for that within the maths.

But the point is that the maths works with or without those assumptions.

So in a sense, it's actually agnostic to that.

And you can conjure an image of a modeler that is pure metaphor,
just talking about synchronization.

However, the language is pretty instructive.

And it does relate back to the idea that lots of systems have and can capture information,
maybe in a very trivial, not cognitive, not conscious-like sense.

But nonetheless, they do capture and synchronize to information,
and therefore they do model and they do inference.

Great. It's the classical, if a tree falls in the forest, does anyone hear it?

Exactly.

If a tree falls, does it make a noise?

Yeah. I mean, I think the answer would be,
it depends on what's synchronizing to the tree, right?

If the tree falls on, okay, this is going to be a pretty grim example, I guess,
but if a tree falls on some sort of small woodland critter, then yes,

I think the tree definitely fell.

Because the statistics of one thing synchronizing to the other one would say,
yeah, pretty unequivocally, the tree just fell.

But if you're not there, you're not synchronizing to the tree.

And so that information has no meaning to you whatsoever.

So it is very much about, as you've written,
modeler-dependent information in a very mathematically rigorous way.

Really powerful and also connects to a lot of qualitative and contemporary ideas.

So it's really exciting to think about where and how Bayesian mechanics can be applied.

This brings us, as we head into the z-score of plus one,
downhill on the Gaussian of the dot one, into the paper itself.

So we'll pause.

Ali and Jakob, if you have any questions.

Otherwise, we're going to go to the roadmap
and have some overview notes on the paper and its structure.

Does that sound good?

All right.

So pardon the meme.

But here's the roadmap of the paper, the sections and subsections.

So maybe just a general comment.

What did you include in the paper?

What did you not include in the paper?

Why did you structure it?

How you structured it?

Yeah, absolutely.

So the structure was kind of meant to reflect, I think, an unmet need in the literature, which is both a single place for all of these results to be reviewed.

And then once that review has been done, an application of all those results.

So organize everything and then use it in a kind of insightful way.

And having all the rigorous stuff in the beginning makes the application make sense.

And having the application makes or provides context for all the very rigorous stuff at the beginning.

So to do that, it's kind of split almost down the middle.

So you have the sections one through four, all the foundational stuff.

And then sections five, six and seven are the actual applications.

So I began with a very brief introduction, just motivating precisely this.

Here are the questions that we're going to answer.

Here's how we're going to answer them.

Here's why it even matters.

In section two, I just very briefly review classical physics to set the stage for some of the stuff that we've already discussed.

Then I talk about Bayesian mechanics itself.

And 2B through 2D are fairly long because this sets up all the basic machinery.

We begin with, I believe, to begin with, we talk about what does it mean for Bayesian mechanics to be like classical mechanics?

What are the basics of it?

What is variational free energy?

What is surprisal?

What is least surprisal?

And then in C and D, these basic ideas can be used to figure out, okay, what does it mean then to minimize surprisal?

How does that matter to Bayesian inference?

Physics of beliefs, physics by beliefs is then inverting the story and talking about,

okay, so we know now that systems, when they minimize their free energy, or minimize their surprisal, they synchronize to their environment.

So you can write this nice story about parameters.

Can you do it the other way around?

Can you begin from the, no, I think I gave that in the wrong direction.

So when systems synchronize, they minimize their free energy.

So you can read this as Bayesian inference.

So then I ask, okay, can you do this the other way around?

Can you begin from Bayesian inference?

And can you get the dynamics of parameters?

Which is just kind of flipping the story, but it involves a bit of work to actually make it follow through.

In the end, however, it does.

And the relationship between the two, so I say physics of beliefs,

because one is taking the motion of parameters and relating it to the motion on statistical manifolds, Bayesian inference.

Physics by beliefs is saying, okay, if we begin with some information theoretic idea of Bayesian inference, do we get back the mechanics of systems carrying those beliefs?

That relies on some work that I did earlier this year,

which espoused a kind of duality between the free energy principle and maximum entropy.

So using that as a kind of technology, you can flip the story and say,

okay, if I start from an inferential principle, can I get physical stuff back?

Whereas the FEP is, I would think of it more as, if I start with physical stuff, can I then understand the beliefs?

They are two sides of the same coin.

And on the nose, it's a very trivial symmetry between the two.

Of course, it takes a lot more work to make it formally make sense, but in the end, it does.

So that's section two.

Section three moves us into the first idea of actual results in the paper is to say,

okay, so if I have a classical system doing Bayesian inference or approximate Bayesian inference, what does that look like?

And does it look like the free energy principle?

And then conversely, if I have an inferential principle, or if I have an informational physics,

can I use maxent as the dual to the FEP to get back the physical mechanics of this classical object?

And in both cases, the answer is yes.

And at the end of that section, I give an equation that does precisely that.

It takes the idea of mode matching and says, okay, here is how probabilities behave when you do mode matching.

And conversely, here is how modes behave when you minimize the free energy of those probabilities.

So in section four, I then talk about, okay, does this actually make sense in a physical setting?

Does it tell us anything insightful?

Or is it just a tautology that if you build in a classical mode and then take the limit to the mode, then you get classical physics?

If you phrase it just like that, it doesn't seem very remarkable.

But for the reasons that I described about path integrals and the classical limit of the path integral, it becomes a little bit more subtle.

And so we can actually derive something kind of interesting from the idea that classical stuff is the mode of quantum stuff.

So in the classical limit of quantum mechanics, systems that do approximate Bayesian inference exhibit

classical physics.

I think that's really the payoff of that section.

And so it kind of justifies the rest of the set of questions that I ask,

because it becomes not just a worked example, but it does say something interesting about physics more broadly, I think.

Now we get into the application section.

So once that's done, we cool off a little bit and we just do an example of what does it mean to do mode matching.

This is simple enough to formulate.

And you can see the, in this case, the mode.

So mode matching is about stationary modes.

And it makes sense to talk about the classical physics of something at rest.

Now things that are at rest don't move.

So synchronization in this case is matching a mode to a stationary environment.

When a force is not acting on you, you don't move.

And this requires very little maths.

And at the end, one of Newton's other laws pops out that objects in motion stay in motion.

Objects at rest stay at rest.

So you get that back from section five.

From section six, you can talk about modes that move.

Well, you can talk about systems that move.

And there's actually a finer distinction in there that I precipitated just saying, which is in one case, you can talk about a system that moves towards a mode.

So a system that tracks a mode, but eventually finishes in a stationary place.

And you can also talk about a system that chases a mode.

So we call this infinite mode tracking, or what you could think of this as mode chasing.

You could think of this as a primitive form of path tracking.

It's just the idea that things that go around in a circle,

you can look at it one of two ways.

You can look at it as an object.

And the example I use is satellite motion.

So the classical physics of celestial bodies, say, one thing orbiting around another.

You can either talk about this as a mode that is being tracked, that is constantly in motion.

So you get infinite mode tracking.

Or you can talk about this as path tracking,

where the path is the circular path around the object.

And what we track is just that path.

And that is what is in 7a.

So I say in 6b, you can talk about this example as infinite mode tracking,
but it's kind of ugly.

It doesn't work very well.

So this justifies going to path tracking instead,
which is a much more elegant picture.

Conceptually, it makes a lot more sense.

And one of the problems specifically with 6b is that the idea of minimizing the surprisal of a mode
becomes kind of difficult,
because this system is non-stationary.

The mode is constantly moving.

And so the surprisal is constantly fluctuating.

But when you talk about the surprisal over paths,
it's much easier.

You can say, okay, we are actually genuinely minimizing the surprisal of paths,
because we are picking the path of least surprisal.

One thing that I didn't include in this section,
which is possibly interesting,
and I don't know, one of the nice things about preprints
is they are to some extent living documents.

So I could come back and add this in if I was so inclined.

In fact, I might.

Is the idea of moving centers of mass.

So path tracking combined with some kind of moving frame of reference.

The reason why I didn't include this is because the maths for that is not quite worked out yet.

But since then,

Lance D'Costa and others,

in a paper that is probably about to be released,

it's co-authored between myself,

Lance D'Costa,

Carl Friston,

Thomas Parr,

Connor Hines,

and Greg Pavliotis.

So all of the big mathematical minds behind the last few years of the FEP.

I think I named all the names.

If I forgot anyone,

then you'll have to excuse me.

There's some work in that preprint,
or that paper that's about to be preprinted,
that formulates path tracking in the context of moving frames of reference.
So not just moving modes along the path,
but also moving paths themselves.
So one extra kind of layer of generalization
is now you can talk about,
okay, what about when the path that I'm tracking
is itself moving?
And there is probably the machinery to handle that
in this paper that's about to be released.
Again, it's Lance, it's Carl,
it's Greg, Thomas, Connor, and myself.
So keep an eye out for that,
because there will be something about
moving frames of reference in that paper.
And I may decide to go back and add something like,
if I'm orbiting a body
that is itself in motion.
So if the center of mass that I'm orbiting
is itself orbiting around something else,
what does that look like
from the perspective of the thing in orbit?
That's an interesting case.
It's something that I didn't include
because it was still,
I think there wasn't an elegant way forward with that.
Now there probably is.
But carrying on,
7b is now relating
the idea of path tracking.
So the idea of a surprisal over paths,
relating that to the path integral approach
to Bayesian mechanics
and extracting some insights
from what it means to be a path integral.
And one of the things that one is allowed to do
in that context

is to then develop
a story about classical chaos.
And I think
we have
probably some slides later about that.
That deserves a much longer conversation.
But the idea of that section is to say,
okay, so we've done all this
arguably pretty trivial stuff.
Now, what about interesting systems
like systems that exhibit
chaos?
Can we actually describe those things
using Bayesian mechanics?
And the answer is we can.
And it comes out of the path integral description
in the nice way that
we have gestured at
with the name G-theory.

Nice.

Thank you.

Jakob?

I was waiting to ask this a bit later on,
but since you mentioned reference frames,
I'm wondering
what different sorts of reference frame
Bayesian mechanics deals with,
especially in connection to
recovering classical mechanics
from Bayesian mechanics.

The way I understand right now,
reference frame is basically imposed
by the Markov blanket.

So when we're talking about
the tree,
the tree branch performing inference
over its environment
and being shaken by

the incoming wind,
that's from the
kind of,
from the reference frame
of the tree.

How would that be different
if we considered

a,
our own reference frame
looking at the,
looking at the tree?

Would that,
could that still have a formulation
within,
within Bayesian mechanics?

And also,
how could this relate to
multi-agent systems
where we have
the classic example
that comes to mind
is clock synchronization
where clocks,
pendulum clocks initially
start out of phase,
but over time

go,
again,
in phase?

I think there's
a lot of different ways
to partition that system.

One clock
is performing inference
over the other,
is performing inference
over the other,
but then maybe

we can also treat it
as a,
as a coupled system
altogether
where we consider
both at the same time.
So,
I'm thinking
there may,
might be
some equivalent
of like a non-inertial
reference frame
in Bayesian mechanics
versus an inertial one
given by the Markov blanket
on that specific system.

So,
we'd be interested
to hear your thoughts
on that.

Yeah,
that's also
a good question.

So,
one of the things
that you can begin
to talk about
in this setting
is that
to some extent
the precise placement
of a Markov blanket
is a bit arbitrary.

And this is,
this kind of flexibility
is actually a good thing
because you do want

to be able to
nest blankets
within blankets.
There's reason
to believe
that that kind
of multi-scale
self-organization,
well,
that,
that multi-scale
patterning behavior,
that,
that multi-scale
separation
and,
and being able
to infer
not only
what my neighbors
are doing,
but being able
to infer
what the scale
above me
is doing.
there's reason
to believe
that that is
where self-organization
comes from
under the free energy principle.
And one of the reasons
to believe that
is work by,
for instance,
Mike Levin
in the context

of the kind of
cognitive nature
of developing cells,
right?

Cells that are

in a kind of
soup of things.

How do they know
to develop
into tissues?

there must be
some kind of notion
of nested blankets,
not only knowing
what am I

in relation

to others,

but what are we

in relation

to a larger scale

other?

So,

that,

that's very important,

nesting blankets.

Another really good

example of this

is in a paper

by Maxwell,

Alec Chance,

some other people

that I don't remember

offhand,

so I apologize,

called neural

and phenotypic

representations

under the free energy principle.

And it's exactly
the same sort of thing
if you have a soup
and they know
something about themselves
and also something
about something
at a higher level,
then you get
self-organization.

So it's,
it's not just about
kind of
self-organization
at the constituent level,
but it's knowing
something about
the pattern
that one
is organizing
into.

And,
and these are
two good examples
of that.

There are lots
of others.
things.

So,
so one imagines,
you know,
the,
the arbitrariness
of the Markov blanket
is important
because it allows
you to nest
things.

More generally,
it allows you
to change context.
And,
you know,
the,
the context
of information
is kind of
observer dependent.

So,
in a way,
you do want
to be able
to introduce
kind of arbitrary
frames,
and get
different kinds
of context
for,
for a given
observation set,
right?

You want to be able
to reassign
self and other
and still get
approximate Bayesian
inference.

This is what
we would think
of as a
gauge theory.

It's a symmetry
at the level
of the system
that is

not reflected
in the
physics of
the system,
by which I
mean,
you can make
free changes
at the level
of the system
that don't
change the
physics of
the system.
So,
in this case,
it's the idea
that we can
change a
reference frame,
we can change
the context
of an inference,
but it shouldn't
change the idea
that systems
do approximate
Bayesian inference,
much in the same
way that we can
change the way
we label coordinates
in general
relativity,
but it shouldn't
change the idea
that gravitation
is the curvature

of space-time
in some suitable
sense that has
to do with metric
tensors that I
won't talk too
much about.
So,
something like that
is absolutely,
I think,
not only
kind of useful,
but is absolutely
essential
to not only
making Bayesian
mechanics make sense,
it would be
kind of suspicious
if inference
just fell apart,
if you redrew
the boundaries,
especially given
our conversation
about the
scale-friendliness
of the FEP.
It should
apply and
reapply,
but also that
application and
reapplication and
contextualizing
inferences within
inferences is

kind of,
right now,
what is reflected
in the literature
when we really
think about,
okay,
how does
self-organization
work?
in these
situations?
And how does
individuation work
in these situations?
Or how are
patterns formed?
And I think
there's lots of
work in this
direction about
reference frames
and the
contextuality of
information outside
the FEP as well.
And I happen to
be familiar with
very specific
literature about
this in category
theory,
but I'm sure
it's more general
than that.
So you can
talk about
how does

context shape
an observation.

So,
yes,
I think,
yeah,
on the subject
of reference
frames and
synchronization,
it's important
to be able to
redraw
boundaries,
so take new
frames of
reference,
and still get
inference.

In the case of
synchronizing
pendulums,
you can imagine,
okay,
maybe the
pendulums do
inference on
each other,
or maybe they
do inference
on the,
kind of,
if they're on
some sort of
table or
moving beam,
maybe they do
inference on the

beam,
and they
synchronize with
the beam,
and therefore
they synchronize
with each other.

That's one frame.

The other frame
is one kind of
looking at the
rhythmic motions
of the other
one,
and just
matching up
like that.

In both
cases,
you get
inference.

We probably
prefer one
reference frame
or another,
but both are
in principle
mathematically
valid,
even if they
tell kind of
different physical
stories about
what's being
synchronized to.

And the
example of
half-tracking

in a moving
frame of
reference is
interesting as
well,
because if you
think about
something like
the moon
orbiting the
earth,
you can zoom
way out,
and you can
look at it
as the moon
spiraling around
the earth as
the earth goes
around the
sun,
but you can
also pick a
reference frame
where the earth
is stationary,
and the moon
is just orbiting
around the
earth.

So this is
the relativity
of motion,
right,
or if you
want to
imagine sitting
in a train,

it looks like

the,

outside is

moving,

and you're

stationary.

So that

would be

mode matching,

you're not

moving.

But from the

perspective of

the outside,

the train has

just gone

past pretty

quickly,

and you're

in the train,

and so you're

moving,

but from your

perspective,

you're not.

So the

relativity of

what is being

matched to

is an important

consideration for

those reasons,

just to produce a

consistent theory,

we have to

acknowledge that

we can freely

assign these

inferences.

The important aspect is that we get the same inferences back, or at least we get inference.

They might not be the same inferences if you wrote them down, they might be in different coordinates, so to speak, but at the end of the day, you should still get Bayesian inference.

Now, I don't remember your second question, so you'll have to ask it again.

I'm actually not aware if there was a second question, but I do have a follow-up question to your answer.

Since you mentioned nested Markov blankets, I'm just

wondering whether
in the
formalism of
Bayesian
mechanics, we
can also
admit a
kind of
dynamics on
the topological
structure of
the blanket
itself, because
in the case
of cells
forming other
layers,
intuitively,
it seems to
me that it's
a kind of
discrete way
of thinking
about how
we're adding
an extra
layer of the
Markov
blanket, and
maybe the
Markov
blanket
initially
admits
some
non-integer
index
connecting to

your prior
work on the
weak Markov
blankets, and
then it
evolves over
time to
form into
actually
statistically
significant
Markov
blanket.

But perhaps
there might
be also a
way for
having just
one Markov
blanket that
evolves through
time, and
that might
also change
the
inferences.

So I'm
wondering
whether
mechanics,
whether the
dynamics of
particular
blankets can
also be
taken into
account in
the path

tracking
formulation or
any other
one within
approximate
Bayesian
inference.

Yeah, so
there are a
couple of
interesting
things there.

For one
thing, the
idea that
perspective
matters, the
idea that
you pick a
reference
frame and
carve up
your
environment
that way,
does kind
of go back
to this
question of
how do we
understand
blankets
forming in
developing
cells, say.
And the
idea would
be that

blankets

have

characteristic

scales.

The reason

why we

see something

like a

stone in

nature and

not a

lattice of

crystalline

molecules is

because at

our level of

observation,

the blankets

that allow

these molecules

to be

self-organized

is completely

meaningless.

But what is

interesting to

our level of

observation, our

kind of

canonical

reference

frame, is

the stone

itself, the

blanket around

the stone

itself.

It's the

stone's

blanket.

It's not

the blankets

that make

the molecules

that make

the stone.

So, yeah,

you know,

eventually we

would want a

much more

complete theory

of individuation

that covers

not just

dynamical

blankets, but

nested blankets

blankets with

perspective and

things of

that nature.

The

perspective-based

idea is an

interesting one

to me because

it allows you

to talk

about things

like sparse

coupling at

certain levels.

And if we

acknowledge that

the Markov

blanket is
coming to a
blanket store
near you,
yeah, exactly.
If we
acknowledge that
the blanket is
somewhat arbitrary,
then we're free
to say that
any system
with enough
couplings has
a blanket at
some scale
because the
kind of
multi-scale
systems in
physics that
we're interested
in at some
scale are
sparsely
coupled,
probably not
every scale,
and certainly
within a
blanket,
there's not
sparse coupling
given a
particular reference
frame because
given the
reference frame

of myself
looking at
the stone,
looking at the
stone's blanket,
within that
blanket,
I would want
to see the
stone's
molecules
cohere in
a way that
is not
sparsely
coupled,
right?
Otherwise,
I wouldn't
see things,
but then at
the level of
the stone,
it is
sparsely
coupled.
So this
kind of
arbitrariness
allows us
to actually
do the
carving up
in a very
interesting
way.
As for
dynamical

systems
perspectives
on Markov
blankets,
I think
this is
also
possible.
I am
working on
something
right now
that takes
the whole
story about
physical
boundaries
and
conditional
independence
and writes
it in
terms of
random
variables.
You have
a whole
suite of
possible
blanket
states,
and you
can look
at blankets
flickering
on and
off,
or blankets

that occupy
different
states over
time.

This is
not nested
yet,
but it's
one step
towards that
direction
where you
can think
about how
the blanket
at a higher
level turns
on or off
given the
dynamics of
its constituents,
whether they're
cohering or
not.

So yeah,
maybe that
answers your
question.

this is a
great topic
to chill
on for a
few more
minutes in
our final
bit of
the dot
one.

There's
really a lot
here.
I'll just
give a few
dots,
and then
Ali,
happy to
hear your
perspective
as well.
You even
discussed it
as scale
friendly,
but it's
almost like
a one and
a half
layer model,
and that is
actually preceded
by even
simple
Bayesian
hidden state
observed
variable
distinctions,
expectation
maximization,
just partially
observable
Bayesian
models in
general,
under an

interpretation
where
observables are
happening at
a given
scale of a
system of
interest,
and a
hidden
phenomena
or a
hidden
cause is
at a
higher
level of
organization,
connects
the sort
of
multi-scale
systems
approach
with a
self-organization
approach,
and we've
talked many
times about
that lateral
and the
vertical,
there's the
nestmates
interacting
laterally,
and then

there's the
real or
as if
colony
level
organization.
Another
point I
think it's
really
interesting to
pull out
and explore
is you
discussed how
important it
is to be
able to
redraw
blankets
and still
get
inference.
It made
me feel
that there
has to
be some
kind of
a locally
distributed
local
sense-making
process
such that
potentially a
vast range
of partitionings

can also
remain valid.

Like,
you can
think of
different
metaphors
there,
and that
potentially
the blanket
provides
exactly that
partitioning.

Surprise,
surprise.

And then
moving towards
all these
features
that had
been in
the
spectator
seats at
the math
conference,
dynamical,
nested,
multi-perspective
systems,
and the
way that
some of
these
formalisms
are pointing
towards an

integration
that's
quite far
out in
some ways,
yet also
may bring
an intuitiveness
to a lot
of the
ways of
talking about
different
physical
systems,
including
the physics
of cognition.
and as
far as
nature at
the
joints,
perspectives
would be
one,
not the
only one,
but one
of those
joints or
sparse
connectivities
or factorizabilities,
various ways of
thinking about it,
but multiple
entities

engaging in a
multi-perspectival,
multi-scale
inference
question,
it's like
two people
looking at
the boat,
they might
disagree on
how to
partition
different parts
of the
boat or
the boat
from the
water,
but at
least their
perspectives
and their
communication
interface
with each
other and
their observation
inference
interfaces with
the boat
could be
specified.
And so
then it
could be
like my
map of

me,
my map
of you,
my map
of the
boat,
your map
of you,
your map
of me,
your map
of the
boat.

And then
there's a
conversation
with all
six of
those
features
instead of
us only
looking at
the boat
and you
did it
one way,
I did it
differently,
and that's
the end
of the
discussion.

So I
think it's
going to
open up
degrees of

freedom
in
perspective
harmonizing
that
wouldn't
be
classically
understood.
And that'll
be seen
as extremely
nonlinear
and
unconventional
behaviors.
And it
just remains
to be seen
what that
is.
I agree.
I'll take
this opportunity
to point
out there
are other
people in
the
Versys lab
who are
thinking about
that kind
of question.
How does
collaboration
and information
synthesis

and kind
of agreement
or goal
emergence,
how do all
these things
fit into
Bayesian
mechanics?
So people
like Maxwell
and
Maholt
are thinking
about if I
write down
these kinds
of problems,
is there a
way to talk
about agreement
as it
emerges
through
collective
discussions or
collaboration?
and it
is likely
something like
this.
It's agreeing
on a
reference
frame for
our
collective
observations

and agreeing
on not
just sense
making,
but it's
agreeing on
how to make
sense of
something.
So there's a
kind of two-step
process.
I learn
something,
figure it out,
I tell you.
You have your
own picture of
what this thing
looks like.
We agree on
a picture that
serves us
best.
We agree
on a
best reference
frame.
Yes.
Agreeing on
the sense
made is
deciding what
to order
after the
dish has
been cooked.
Exactly.

We need to
have a
conversation
about the
recipe and
so many
other things.

Ali?

Well,

I believe
this whole
argument
around

Markov

Blanket

and

especially

the

Nested

Markov

Blanket

or the

dynamic

nature

of them

can

potentially

be

connected

to

somehow

the

opposing

stances,

one of

which is

the

preformism

or
preformationism.

The other
one is
the
progressive
differentiation
mechanism
which
in the
preformism
some people
believed
that
the
topology
of the
whole
state
space
manifold
is
completely
preformed.

We just
don't have
enough data
to model
that accurately,
model the
state space
accurately.

But
for the
advocates
of the
progressive
differentiation

viewpoint,
they, on the
other hand,
believe that
it literally
goes, I
mean, the
state space
manifold
literally
undergoes
lots of
transformations
according to
some
inflection
points or
some
singularities
and it's
not
preformed
at all
and
the
dynamic
nature
of
the
state
space
manifold
is
something
inherent
to that.
It's
not

because
of
our
lack
of
enough
data
to
model
that.
So
I'm
not
sure
how
much
you
agree
with
that
viewpoint
something
that's
Yeah,
I think
sure,
I think
the
kind of
drawing
and
redrawing
of
Markov
blankets
is
probably
a point

in support
of the
idea
of
progressive
differentiation
because
at one
level
we think
of
drawing
and
redrawing
as
development
or
yes,
development
and one
of the
things
that
drives
development
is
something
like
fitness
pressure.
So
fitness
changes
the
constraints
on
what
sort

of
patterns
are
available
to
you,
what
ought
you
organize
into.
And
these
are not
things
that we
know
a priori
at the
level
of the
system.
It
may be
possible
that if
you write
down
a whole
kind
of a
world
model
of the
entire
universe
from
the

very
first
point
in
time
after
the
Big
Bang
or
maybe
even
at
the
instant
at
which
the
singularity
expanded
and you
had enough
computational
power to
trace it
all through
that you
could get
to
something
like
what is
it
preformationism
but that
strikes me
quite a
lot as

a kind
of
super
determinism
or
determinism
argument.

It sounds
a lot
like
some
kind
of
argument
that

there is
no
uncertainty
in how
you will
develop.

There is
just
uncertainty
in
what
you're
aware

of
as to
your
development
or
some
kind
of
hidden
variables

theory
that
says
there is
something
mediating
the way
your
state
space
looks
you
just
don't
know
it
and
so
you
have
the
illusion
of
free
will
so
this
becomes
a bit
of a
philosophical
question
I
myself
don't
really
buy
into

determinism
so
I
would
be
tempted
to
say
that
the
Markov
blanket
story
especially
dynamic
blankets
or
however
you
want
to
call
them
are
to
do
with
progressive
differentiation
as
we
live
our
lives
we
draw
and
redraw

individuate
ourselves
from
other
systems
and
the
environment
and
this
is
something
that
we
are
in
control
of
certainly
locally
in the
state
space
we're
in
control
of
it
whether
there
is
some
hidden
variable
guiding
our
own
actions

is
something
that
is
I think
strictly
speaking
a
different
question
the
nice
thing
about
it
is
whether
it's
true
or
not
it
doesn't
break
the
locality
of
the
Markov
blanket
construction
so
we
get
this
idea
of
locally

progressive
differentiation
as a
Markov
blanket
and
then
when
we
zoom
way
out
it
may
be
a
different
story
but
certainly
locally
it
works
as
a
dynamic
process
well
we
almost
got
to
the
paper
this
was
an
awesome

discussion

so

Ali

and

Jakob

if

you

would

like

to

have

any

closing

remarks

well

I

just

want

to

say

that

I

truly

enjoyed

our

discussion

today

and

I

very

much

look

forward

to

that

too

because

I

have
a
bunch
of
questions
that
I
didn't
have
a
chance
to
ask
about
some
of
the
sections
of
the
paper
so
I
just
wanted
to
thank
Dalton
for
his
time
and
also
for
Jakob
and
Daniel
for

organizing
these
extremely
fruitful
and useful
discussion
sessions
and live
streams
so
yeah
I'm
very much
looking
forward
to
the
next
one
thank
you
Jakob
yeah
this
was
an
amazing
amazing
dot
one
thanks
thanks
a lot
Dalton
for
all
the
great

answers
and
overviews
of
course
Ali
and
Daniel
for
doing
bulk
of
the
work
on
the
slides
and
yeah
looking
forward
to
the
dot
two
Dalton
any
last
thoughts
no
just
thank
you
for
having
me
I
did

really
enjoy
it
and
thank
you
of
course
for
all
the
background
work
that
you
put
in
it's
as
I
said
it's
very
flattering
to
see
such
detailed
working
through
of
my
work
so
thank
you
for
that

and
I
am
looking
forward
to
the
next
episode
excellent
well
next
week
us
and
anyone
else
who'd
like
to
join
just
get
in
touch
or
submit
any
questions
or
topics
so
thank
you
fellows
see
you
later

okay

cheers

thanks

to